8/10/26

EXPERIMENT - 12 (a)

END-END COMMUNICATION AT TRANSPORT LAYER

AIM: Implement echo client server using TCP/UDP

TCP:

```
import socket
import threading
def start_server():
    server = socket.socket (socket.AF_INET,
socket.SOCK_STREAM)
    server.bind ('127.0.0.1', 5000)
    server.listen(1)
    print ("[SERVER] listening on 127.0.0.1:5000")

    conn, addr = server.accept()
    print (f"[SERVER] ~~listen~~ connected by
                {addr}")

while true:
    data = conn.recv(1024).decode()
    if not data or data.lower() == 'quit':
        print("[SERVER] connection closed
    by client")
        break

    conn.close()
    server.close()

def start_client():
    client = socket.socket (socket.AF_INET,
socket.SOCK_STREAM)
```

```
    client.connect(('127.0.0.1', 5000))
    print("[CLIENT] Connected to server")
    while True:
        msg = input("Enter message")
        client.send(msg.encode())
        if msg.lower() == 'quit':
            break
    client.close()

if __name__ == "__main__":
    server_thread = threading.Thread(
    target = start_server, daemon = True)
    server_thread.start()
    start_client()
```

O/P:

```
[SERVER] listening on 127.0.0.1:5000 [CLIENT]
connected to server
[SERVER] connected by ('127.0.0.1', 53458)
Enter message:
 hello
[SERVER] Received: Hello
[SERVER] Server Received: hello
Enter message:
[SERVER] listening on 127.0.0.1:5000 [CLIENT]
connected to server
[SERVER] connected by ('127.0.0.1', 60828)
Enter message:
   HI
```

[Server] received: HI

[Server Response] Server received: HI

Enter message: quit

[Server] connection closed by client.

UDP:

```
import socket
import threading
def udp_server():
    S = socket.socket(socket.AF_INET, socket.
    SOCK_DGRAM)

    S.bind(("127.0.0.1", 2000))
    while True:
        data, addr = S.recvfrom(1024)
        msg = data.decode()
        if msg.lower() == 'quit':
            print(f"[SERVER] client {addr}
            ended the connection")
                break
    S.close()
    print("[SERVER] stopped.")
def udp_client():
    C = socket.socket(socket.AF_INET, socket.
    SOCK_DGRAM)
    while True:
        msg = input("Enter message")
        C.sendto(msg.encode(), server)
        print("Echo from server:", data.decod
    C.close()
    print("[client] stopped")
```

```
if name == "__main__":
    threading.Thread(target=udp)
    rt().udp_client()
```

O/P:

Enter message : [SERVER] Running on 127.0.0.1: 12000

[SERVER] Received from ('127.0.0.1', 49670): hi

Echo from server: hi

Enter message: hello

Echo from server: hello

Enter message: how are you

[SERVER] Received from ('127.0.0.1', 49670): how are you

Echo from server: how are you

Enter message: quit

[CLIENT] closing connection

[CLIENT] stopped [SERVER] stopped.

Result:

Thus the end to end communication has been executed